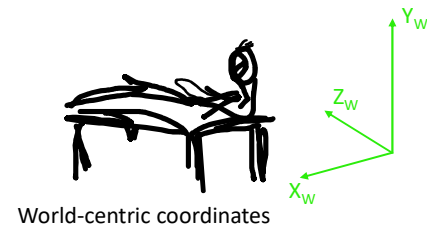
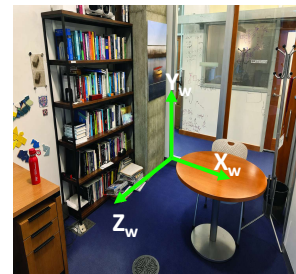


Last Time

- If we have the perspective camera matrix P
 - the null space of P corresponds to the origin of the camera in the world coordinate system



Assume you know the camera matrix
Where was this picture taken from?



1

Aside: Weak Perspective Camera



- Used often in the literature
- Scene is assumed to have approximately constant depth = average depth
 - If the variation in depth (Z coordinate) of different points in the scene is negligible compared to the average depth Z' of the scene, then

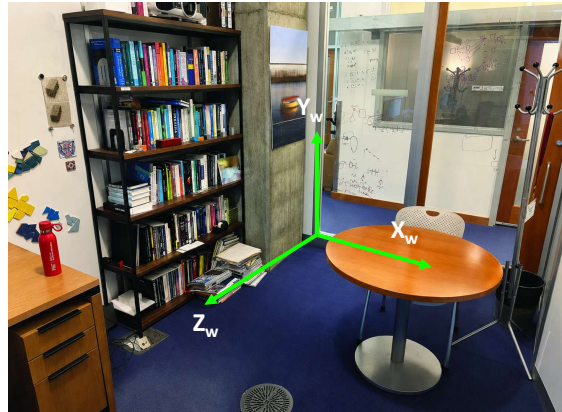
$$x = f \frac{X}{Z + \delta} = f \frac{X}{Z} \left(1 - \delta/Z + (\delta/Z)^2 \right) \approx f \frac{X}{Z},$$

$$y \approx f \frac{Y}{Z}$$

2

Agenda

- How to get hold of the camera matrix?
- This topic is called camera calibration



3

Agenda: Camera Calibration

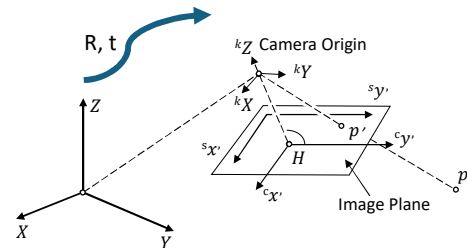
- Calibrated Cameras
 - Why 
 - What
 - How to calibrate
 - Method 1
 - Method 2
 - Method 3



4

Camera Calibration: What

- $P = K[R \ t]$ is the camera matrix
- Goal: Get P
 - K : camera intrinsics
 - R, t : camera extrinsics



$$K = \begin{bmatrix} \alpha_x & s & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$

$$x_{3 \times 1} = KR \left[I_{3 \times 3} \mid -\hat{C} \right] X_w = P X_w$$

5

Camera Calibration: Why

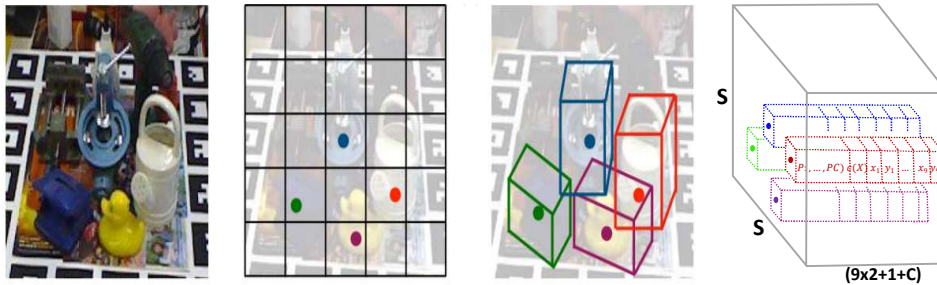
- Given pictures from a calibrated camera (or a set of pictures from calibrated cameras), we can reconstruct parts of the underlying 3D scene and perform augmented reality
- We can estimate various dimensions in 3D given just the 2D image(s)
- We can estimate camera properties such as the focal length



6

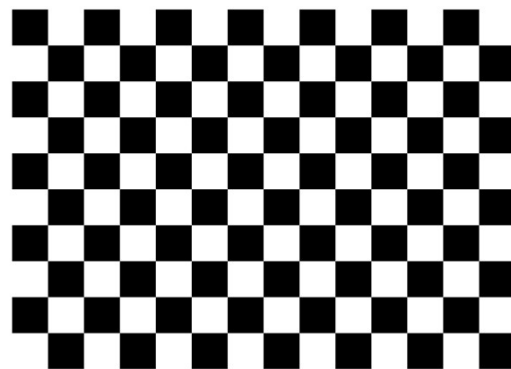
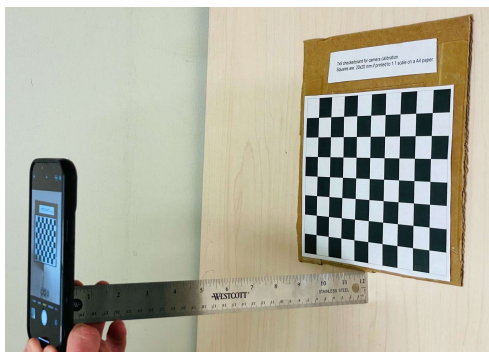
Geometric Camera Calibration: Why

- To get to (e.g.) 6D pose (in camera view) we need to obtain the parameters of the (unknown) camera
 - We need to go from 2D to 3D



7

Simple Method to get K



8

Simple Method to get K

$$\alpha_x = fN/w$$

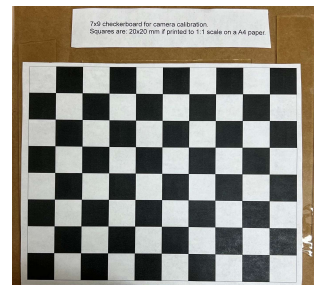
$$\alpha_x \approx ZL/W$$

$$K = \begin{bmatrix} \alpha_x & s & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$

- **Task: Verify with manufacturer website**
- Note: If, for example, manufacturer says 35 mm equivalent focal length is 26mm
 - Then $f = 26/36 \cdot \text{sensor width}$
 - This is because a physical 36mm camera has sensor size of 36x24

- f : focal length
- N : Number of pixels in the horizontal direction
- w : sensor width

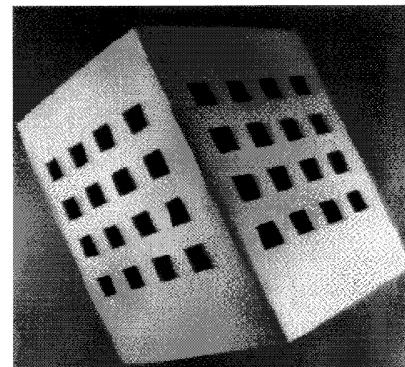
- L : width of black square in pixels
- W : width of black square in cm
- Z : distance of capture in cm



9

Camera Calibration: How?

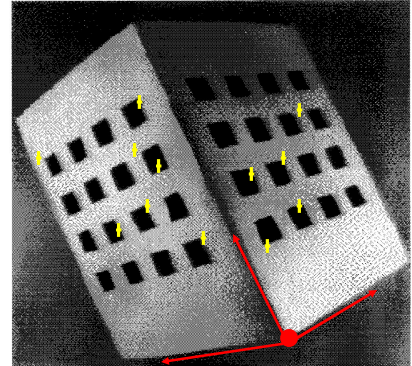
- Take photographs of an object whose geometry is known and which will produce salient feature points in an image
- Example: checkerboard cube
- Such an object may be termed as a calibration object



10

Camera Calibration: How?

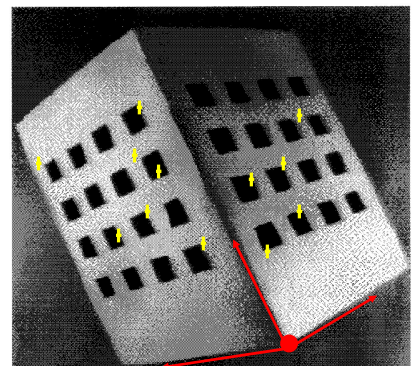
- Measure 3D coordinates of various junction points (marked yellow) of such a calibration object.
- Express the coordinates with respect to an origin O that is chosen



12

Camera Calibration: How?

- Capture an image
- Measure the coordinates of the **pixel** locations of the same junction points
 - Thus for each of the N junction points, we have the pixel coordinates as well as the world coordinates.
- Recall DLT
 - From this, derive the elements of the P matrix and then the extrinsic and intrinsic parameters of the camera.



13

$$\begin{aligned}
 & \checkmark X_1 = \begin{pmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \end{pmatrix} \begin{matrix} X_1 \\ Y_1 \\ Z_1 \\ 1 \end{matrix} \\
 & \checkmark Y_1 = \dots \\
 & \checkmark W_1 = \dots \\
 & P.C \quad \uparrow
 \end{aligned}$$

$$Y_1 = P_{21}X_1 + \dots + \dots$$

$$X_1 = P_{11}X_1 + P_{12}Y_1 + P_{13}Z_1 + P_{14} \quad y = \frac{y_1}{w_1}$$

$$w_1 = P_{31}X_1 + P_{32}Y_1 + P_{33}Z_1 + P_{34}$$

$$\boxed{x} = \frac{x_1}{w_1} = \frac{P_{11}X_1 + \dots + P_{14}}{P_{31}X_1 + \dots + P_{34}}$$

$$\Rightarrow P_{31}xX_1 + P_{32}xY_1 + P_{33}xZ_1 + P_{34}x - P_{11}X_1 - P_{12}Y_1 - P_{13}Z_1 - P_{14} = 0$$

$$\Rightarrow \begin{bmatrix} xX_1 & xY_1 & xZ_1 & x & 0 & 0 & 0 & -X_1 & -Y_1 & -Z_1 & -1 \end{bmatrix}_{1 \times 12}$$

$$\begin{bmatrix} -X_1 & -Y_1 & -Z_1 & -1 & 0 & 0 & 0 & xX_1 & xY_1 & xZ_1 & x \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & -X_1 & -Y_1 & -Z_1 & -1 & yX_1 & yY_1 & yZ_1 & y \end{bmatrix}$$

$$\begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ P_{14} \\ P_{21} \\ P_{22} \\ P_{23} \\ P_{24} \\ P_{31} \\ P_{32} \\ P_{33} \\ P_{34} \end{bmatrix}_{12 \times 1} = 0$$

14

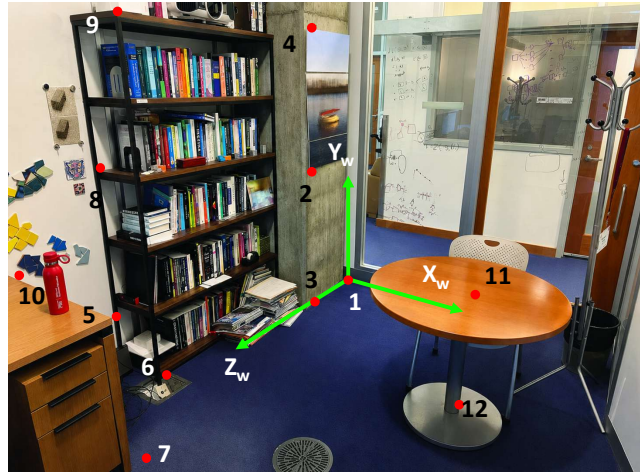
$$A p = 0$$

$$\begin{array}{l}
 P_{12 \times i} \\
 A_{2N \times 12} \\
 \hline
 P \dots \text{upto a constant} \\
 (11)
 \end{array}$$

15

Example

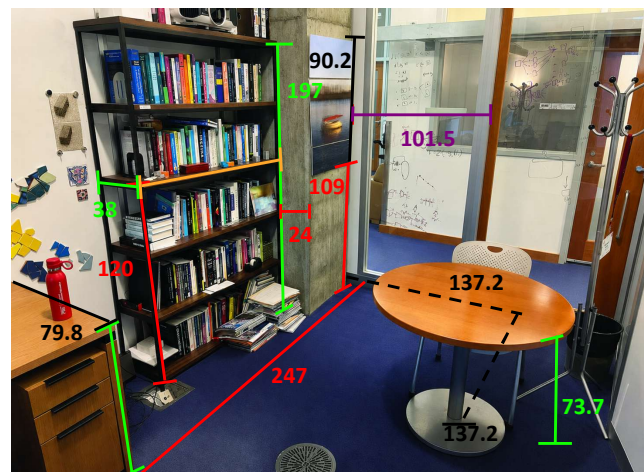
- Set up a world origin
- Identify a few points



16

Example

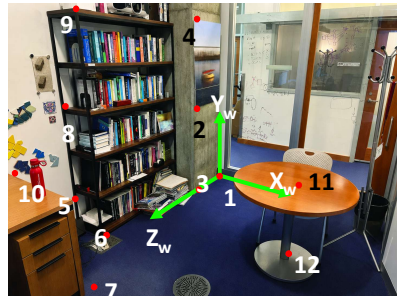
- Set up a world origin
- Identify a few points
- Measure



18

Example

- Set up a world origin
- Identify a few points and measure
- Capture an image
- Use Microsoft Paint or similar to identify world coordinate to pixel coordinate mapping



i	X_i	Y_i	Z_i	x_i	y_i
1	0	0	0	2182	1261
2	0	109	73	1931	1952
3	0	0	73.7	1902	1064
4	0	199.2	73	1917	2855
5	17.8	76.2	247	670	1005
6	-24	0	193.7	979	612
7	17.8	0	247	793	50
8	-62	120	193.7	550	1942
9	-24	197	193.7	692	2963
10	-62	76.2	247	30	1268
11	137.2	73.7	137.2	2938	1137
12	137.2	8.8	137.2	2836	386

19

Example

- Set up a world origin
- Identify a few points and measure
- Capture an image and get pixel values
- Apply DLT
 - Compute SVD
 - Extract right singular vector corresponding to smallest singular value

X	Y	Z	u	v
0	0	0	2182	1261
0	109	73	1931	1952
0	0	73.7	1902	1064
0	199.2	73	1917	2855
17.8	76.2	247	670	1005
-24	0	193.7	979	612

$$\begin{bmatrix} -24 & 0 & 193.7 & 1 & 0 & 0 & 0 & 0 & 23496 & 0 & -189232.3 & -979 \\ 0 & 0 & 0 & 0 & -24 & 0 & 193.7 & 1 & 14688 & 0 & -118564.4 & -612 \end{bmatrix}$$

20

Example

- Extract right singular vector P
- Reshape to form matrix
- Normalize
- Compare

$$P = \begin{bmatrix} -8.1 & -1.8 & -0.2 & 1845.5 \\ -3 & 5.4 & -4.8 & 1262.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This was done with
six random points
using a LLM

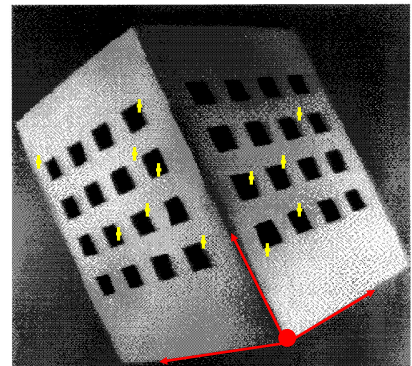
$$P = \begin{bmatrix} -2.2633 \times 10^{-3} & 5.9281 \times 10^{-4} & 2.9948 \times 10^{-3} & -8.6581 \times 10^{-17} \\ 5.4994 \times 10^{-4} & -2.1253 \times 10^{-3} & 1.9055 \times 10^{-3} & -5.0034 \times 10^{-1} \\ -2.8059 \times 10^{-7} & 3.2104 \times 10^{-7} & 7.9507 \times 10^{-7} & -3.9680 \times 10^{-4} \end{bmatrix}$$

$$P \approx \begin{bmatrix} 5.70 & -1.49 & -7.55 & 2182 \\ -1.39 & 5.36 & -4.80 & 1261 \\ 0.000707 & -0.000809 & -0.002004 & 1 \end{bmatrix}$$

21

Camera Matrix

- We got P
- But how do we determine intrinsics and extrinsics?
 - Determine K , camera intrinsics
 - Determine R, t , camera extrinsics
- Recall $P = [KR | -KR \hat{C}]$
 - And focus on the top left 3×3 matrix
 - That matrix is the product of K and R



22

QR Decomposition

$$A = Q R$$

$\begin{matrix} \text{orth} & & \text{up} \\ \swarrow & & \searrow \\ Q & & R \\ \nwarrow & & \nearrow \\ P & = & K & R \end{matrix}$

- Any real invertible $A_{n \times n}$ can be factored as $A = QR$ where Q is an orthogonal matrix and R is an upper triangular matrix with non-zero diagonal

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & 3 & 3 \\ -1 & -1 & 5 \\ 1 & 3 & 7 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 & 4 & 2 \\ 0 & 2 & 8 \\ 0 & 0 & 4 \end{bmatrix}$$

Can we do QR decomposition and claim that the K we want is R ?

$$= \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ 0 & R_{22} & R_{23} \\ 0 & 0 & R_{33} \end{bmatrix}$$

$= QR$

23

RQ for square matrices

- Suppose we know a QR decomposition of A
 - Given $A = QR$, how can we get $A = RQ$?

24